

A Reevaluation of Working Memory Capacity in Children

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A common account of working memory capacity, and its development, incorporates the notion of limited mental resources being shared between processing and storage components. Working memory tasks are assumed to measure this capacity. However, recent data from children's performance at one such task, counting span, implicate forgetting over time as being more important than the capacity for resource-sharing. Three experiments investigated the role of temporal factors in more depth and breadth by varying the retention requirements in working memory tasks while holding constant the overall processing difficulty. Counting span, operation span, and reading span tests in 6- to 11-year-olds were all shown to decline as the retention interval of stimulus items is increased and, across tasks, there were no consistent effects from different concurrent storage loads on processing speed. It is suggested that among children, performance in these tasks does not reflect a trade-off between resources allocated to processing and storage and that caution should be adopted in interpreting working memory span in adult populations. © 1998 Academic Press

Much cognitive activity rests on the ability to maintain temporarily, as well as transform, in-

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formation relevant to a task. Partial products of various cognitive processes must be retained and reintegrated with other computations to generate a useful and veridical representation or a solution to a problem being tackled. Such a perspective highlights the value of theoretical models that can account for transient memorial processes. One popular model is that of "working memory," as advanced by Baddeley (1986; Baddeley & Hitch, 1974; for related development accounts, see Case, 1985; Case, Kurland, & Goldberg, 1982). Within such a view, immediate memory is not seen as a passive repository of events, but as part of an active and integrated cognitive system. It follows from this stance that measures of working memory should reflect the interplay of processing and storage taking place in real-world tasks.

Accordingly, there now exist several tests of

working memory capacity that measure the ability to retain information while simultaneously carrying out ongoing processing activities. Such tests include reading span (Daneman & Carpenter, 1980), (arithmetic) operation span (Turner & Engle, 1989), and counting span (Case et al., 1982), each named after the type of processing activity entailed. While it would be overly optimistic to claim that there is complete consensus over the status of these active, working memory span tasks (e.g., Daneman & Tardif, 1987; Engle, Cantor, & Carullo, 1992; Waters & Caplan, 1996), it is often held to be the case that they have more potential as measures of working memory than their "passive span" counterparts, such as digit span or word span, which lack a substantial processing component (see Daneman, 1995). For example, working memory span tasks are reported to provide better predictions of reading skill than do word span tasks (Daneman & Carpenter, 1980).

The starting point for the present analysis is an influential study on counting span by Case et al. (1982). In this task, an individual is presented with a series of displays, each containing a mixture of targets and distracters. For each display, the subject counts the number of target objects in the array. Immediately after completing the final count, the subject is asked to recall the totals for all the displays. Counting span is measured by increasing the number of displays to find the maximum number of counts that can be made with correct recall of totals. Case et al. (1982) measured counting span among 6- to 12-year-old children and, separately, also measured processing difficulty (as indicated by children's counting speed). Furthermore, adults performed the span task using nonsense names as numbers, a manipulation designed to increase cognitive demand. Across all age groups, Case et al. (1982) found a linear relationship between memory capacity and processing speed, so that older children counted more quickly and obtained higher spans. Adults' span, in counting with an unfamiliar sequence, was reduced to that of 6-year-olds, and their processing speed corresponded to that of the 6-year-olds also. Case et al. (1982) accounted for these results by proposing a model in which there is a limited

capacity pool of resources that can be flexibly allocated to either storage or processing. Thus, these data were accounted for in terms of *resource sharing* or a trade-off between processing and storage demands; individuals with faster processing speed were held to require fewer resources for counting, enabling them to devote more resources to memory operations. Similar resource trade-off interpretations, sometimes specifying resources as activation energy, have been used to explain adult performance on working memory span tasks involving other processing operations (e.g., Daneman & Carpenter, 1980; Just & Carpenter, 1992; Turner & Engle, 1989).

However, while consistent with a resource trade-off account, the data from Case et al. (1982) can be given an alternative interpretation. Task difficulty may impact upon working memory span, not because of a change in the balance of shared resources, but because of the additional processing time required for harder tasks. That is, slower processing extends the time period over which memory items may be forgotten (Towse & Hitch, 1995; see also Halford, Maybery, O'Hare & Grant, 1994). Thus, contrary to conceptions of working memory tests which emphasize resource sharing between processing and storage, Towse and Hitch (1995) argued—at least for children's counting span—that instead of actively combining storage and processing operations, children may alternate between counting during display presentation and storing results at display offset. Given such a *task-switching* strategy, the effect of processing speed on counting span can be explained simply by assuming that stored results undergo forgetting during the processing phase of the task.

Towse and Hitch (1995) based their conclusion on the analysis of counting spans among groups of children between 6 and 10 years, following independent manipulation of task complexity and task duration. The complexity of counting was varied by requiring children to ignore nontarget items either distinct from, or similar to, target squares, a manipulation known to have robust effects on visual attention mechanisms (Treisman & Gelade, 1980). The card

counting time was manipulated by varying the total number of target squares. According to the resource trade-off interpretation, counting span should be sensitive to the intensity of the mental effort of the counting operations, whereas according to the task-switching interpretation, the important factor should be the duration of processing. The results showed that, for each age group, span was a function of the total duration of the counting operations. Complexity of counting had no effect beyond that attributable to count duration. Accordingly, Towse and Hitch (1995) suggested that children do not share resources between processing and storage and that counting span does not measure the amount of working memory resources available for trading between these functions.

The distinction between resource sharing and task-switching is important theoretically since it applies in principle to all working memory span tasks. Given some of the subtleties involved, a brief commentary may be helpful. Towse & Hitch (1995) suggested memory decay over time was responsible for loss of information in the counting span task. However, time itself need not produce degradation in memory, since, for example, the impact of intralist interference may change with time. Either way, storage is independent of concurrent processing load, and processing performance is independent of concurrent storage load. The relationship between processing and storage arises because the time spent in processing affects the amount of forgetting that accrues. At the core of the task-switching model is the notion that working memory span tasks invoke either a serial processing strategy or utilize separate systems dedicated to processing and storage functions.

In contrast, standard resource-sharing models presuppose that there is a dependence between retention and processing systems. This is often instantiated in the proposal that there is some general-purpose system that can perform multiple functions, with a limited ability on the part of this system to service the requirements of different cognitive activities simultaneously. Views differ as to just how general-purpose the system is, and the point at which there is an efficiency cost in combining activities, but at heart is the notion

of mutual competition between processing and storage activities. This might arise from a conventional model with shared resources (Case, 1985; Daneman & Carpenter, 1980; Turner & Engle, 1989) or from a modified position by which the number of basic processing events—the number of “things to do”—affects storage (Case, 1995; Case, Okamoto, Griffin, McKeough, Bleiker, Henderson, & Stephenson, 1996). It should be apparent that the basic tenet of resource sharing is preserved in this modified approach: the mutual influence of processing and retention, albeit by informational cross-talk rather than resource competition.

The present experiments sought to provide empirical evaluations of these ideas, since they lead to different predictions for performance. An experimental paradigm was developed that could be applied across processing task domains, where overall task difficulty was comparable (in terms of the number of basic elements making up the processing phase and from measurements of on-line processing performance) while retention requirements were varied. Thus, we attempted to confirm the importance of temporal factors in counting span, as well as in other working memory span tasks. This was achieved via adaptation of an elegant paradigm used by Cowan, Day, Saults, Keller, Johnson, & Flores (1992) to explore word length effects in standard serial recall situations. To appreciate the present approach, Cowan et al.'s experimental method is briefly considered.

Cowan et al. (1992) evaluated verbal rehearsal and output decay as sources of word length effects (see Baddeley, Thomson, & Buchanan, 1975). The former hypothesis proposes that long stimulus words impair memory performance because of the reduced ability to rehearse items in a time-limited store. The latter hypothesis posits greater forgetting during the output of items (long words take longer to say and hence delay output). To tease these hypotheses apart, Cowan et al. (1992) presented subjects with lists of items comprising long and short words, placed either early or late in the sequence. They found that when the initial recall items involved long words, there was a greater impact on subsequent retrieval efforts

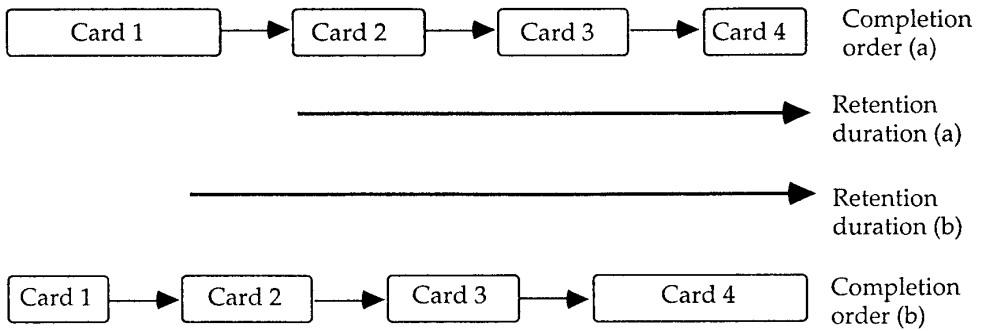


FIG. 1. Schematic representation of the time course when processing a set of four cards with completion order varied. For completion order (a), the initial card takes a relatively long time to complete. For completion order (b), the initial card takes a relatively short time to complete. Arrows indicate the time period during which retention is required, and this shows that order (b) involves additional retention time, even though overall task length is equivalent to order (a).

than was observed when initial recall items involved short words. Cowan et al. interpreted this finding as showing that trace decay at output is important to the word length effect.

This experimental design was adapted in the initial study so as to manipulate the retention duration of items during the *processing* phase of the counting span task. Children were presented with sequences of cards to count, where the array numerosity of the first card was small (and the last card large) or the numerosity of the first card was large (and the last card small). The numerosity of intervening cards was not altered. For example, there might be three cards with four–five–seven squares to count when the final card is large and seven–five–four squares when the final card is small. Thus in both conditions (small–final and large–final), the overall processing demands of array counting should be identical since the same set of cards has to be processed and remembered. Accordingly, in terms of cognitive difficulty of counting, and in terms of the overall concurrent activity, the cards ought to lead to equivalent counting spans.

However, the manipulation of completion order is relevant to the time during which representations are maintained in working memory. Since only count products require retention, when counting the first card there is no memory load. When the final card is being counted, however, all preceding totals must be retained. Therefore, when the final card involves larger arrays (and thereby,

longer counting times), there is an increased likelihood of forgetting before recall commences. This is because the maintenance of memory stimuli must be extended over a greater time period, even though the cognitive processing to complete the task is the same.

This can be seen schematically in Fig. 1, where four cards are processed (counted). For completion order (a), retention of information begins after a relatively large time interval and continues until the final, quick, card has been processed. While the overall processing time in completion order (b) is the same (i.e., starting and terminating at the same point), because the quick and slow cards have been switched, the start of the *retention* requirements begin sooner, and so this completion order produces additional temporal demands on retention mechanisms. It is important to note that here, unlike in Cowan et al. (1992), it is merely the retention requirements of memory items that vary across experimental conditions, while the output characteristics are equivalent.

Thus, the ability to combine the retention of items with the completion of cognitive operations is assessed under relatively low or high temporal demand conditions. To preview the experiments that follow, the cognitive tasks involved counting (to yield counting span), arithmetic sums (to yield operation span), and sentence comprehension and inference (to yield reading span), as these represent the common

types of working memory span test. And in the final experiment, we also incorporated separate assessments of processing speed to explore further the locus of working memory capacity.

By developing a computer-administered version of the span task, we were also able to compare processing speed at the beginning and the end of each trial. This allowed for a further assessment of theoretical models of working memory span. An interesting contrast between the resource sharing and task-switching accounts concerns the effect of different concurrent storage loads on processing performance. Resource sharing accounts predict that processing will suffer interference from concurrent storage load. Processing at the end of the trial (when storage load is maximal) should be impaired relative to processing on the first item (when no storage load is imposed). Task-switching, on the other hand, predicts that processing will be unaffected by concurrent storage load.

All three experiments used children whose ages are within the range studied previously (e.g., Case et al., 1982). This period of development is one in which there are marked changes in strategy use in memory tasks (Bjorklund, 1990; Kail, 1990). We entertained the hypothesis that the way children perform working memory span tasks might also change over this period, from early dependence on a task-switching approach to later predominance of resource sharing. Such a change might have many causes, such as increased task efficiency or maturational changes in ability. In the first instance we were interested only in seeing whether a difference of this sort could be detected using the present methodology.

EXPERIMENT 1

This experiment explored the relevance of processing duration for counting span in children of different ages. It also provided an opportunity to evaluate explanations for performance in terms of resource sharing. Whereas previously, processing difficulty was manipulated while duration was controlled for (Towse & Hitch, 1995), the present experiment manipulates processing duration whilst keeping difficulty (and interference dosage) constant. Ac-

cordingly, a task-switching account predicts that counting spans will vary with retention duration. A resource sharing account, in contrast, predicts that spans will be equivalent since mental effort is matched across experimental conditions. At the same time, these sharing models predict that processing will be impaired when the concurrent memory load is high, while switching models predict no effect.

Method

Subjects and Design

Data were available for 67 children in four age groups attending a Surrey primary school. Age group categories reflected class assignment at the beginning of the academic year. There were 18 6-year-olds (mean age was 6.9, ranging 6.3 to 7.2), 17 7-year-olds (mean age 7.8, ranging 7.3 to 8.1), 19 9-year-olds (mean age 9.9, ranging 9.4 to 10.2), and 13 10-year-olds (mean age 10.5, ranging 10.2 to 11.1). All children completed two counting span assessments with presentation order rotated across subjects. For the purposes of exposition, children's performance on trials where the final card was large (and hence first card small) or where the final card was small (and hence first card large) is termed the effect of final card size, and this formed the within-subjects manipulation.

Count Arrays

The counting span arrays were administered via an Archimedes 3000 microcomputer controlling a 12-in color Acorn monitor. The number of target squares appearing on each card varied between three and eight, with a mean value of 5.5. When small cards were presented, they contained three, four, or five squares to count, with a mean of four, and when large cards were presented, there were six, seven, or eight squares, with a mean of seven. Target squares (approximately 8 mm in length) were displayed in blue, with distinct nontarget triangles (9 mm in length) colored light red. With a small variability, there were twice as many nontarget triangles as there were target squares, and the target-nontarget object positions were intermingled (object proximity was slightly higher

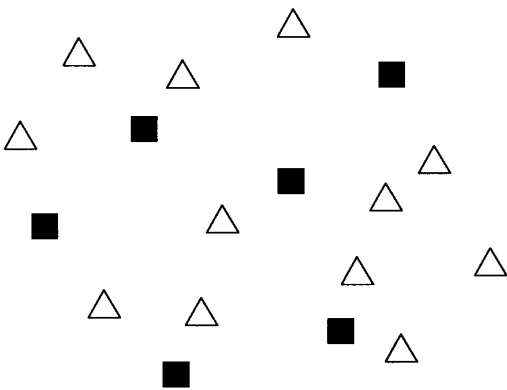


FIG. 2. Example spatial arrangement of targets and non-targets in Experiment 1. Squares were presented on screen in blue, triangles in red.

than for Towse & Hitch, 1995, where physical cards were used, but was roughly comparable). Objects appeared on a gray background, with a yellow border to represent the outline of a card, and the remaining cards to count, if any, shown as partially occluded outlines (see Fig. 2). The computer stored six independent spatial arrangements for each card numerosity (i.e., six different cards for each possible array size). The numerosity of target squares on each trial was fixed, but the selection among spatial arrangements for that card size was determined randomly. This meant that counting the same card numerosities on different trials would likely involve different visual displays, reducing the learning of pattern-answer associations.

In the small-final condition, the first card was a large array and the last card a small array, other cards being selected from both small and large sets. The large-final condition was similar except that the first and last cards were presented in reversed order, so that now the first card was a small array and the last card a large array, with the same intervening array sizes.

Procedure

All children were tested individually by the same experimenter. The task was introduced to children as a memory game. This game was initially explained to children using two laminated cards. Children were instructed to count all the squares on the first card, pointing to each

as they did so, and ignoring triangles on the card. Pointing to the targets resulted in children counting aloud as they worked on each card. They were also told that the game involved remembering each card total, so that these could be recalled after counting all cards. It was emphasized to children that they should not say anything between card counts and that they should try hard to count all the cards correctly.

The span tests started with two count cards. A trial began with a centrally placed “Get ready . . .” signal on the computer monitor, which appeared for a random time between .5 and .75 s, after which the first card was shown. As soon as the child verbally gave a count total to the first card, the experimenter entered this answer into the computer and the next card appeared for counting (to avoid abrupt visual changes, the delay between answer entry and the display of next card was .3 s). After both arrays had been counted, small blank cards were displayed across the screen and the child was cued to recall sequentially the number of squares on each card (using a verbal report). The correct memory response was always the relevant answer produced when counting, not necessarily the actual count total. At the end of the recall phase, visual feedback was provided concerning recall accuracy, together with a warning message if there were multiple errors in counting targets.

The number of cards to count (and hence totals to recall) increased by one each time the

TABLE 1
Counting Times for Cards in Experiment 1

	Age			
	6	7	9	10
Counting processing rate (seconds per counted square)				
First cards	.86	.75	.60	.60
Last cards	.75	.68	.65	.59
Counting time on the last card in the set (in seconds)				
Small card	3.19	2.87	2.67	2.38
Large card	4.92	4.48	4.29	4.03

child correctly remembered the set of totals on two trials (out of a possible three). The third trial was omitted if both preceding trials were remembered correctly. When more than one set of cards among all three was recalled incorrectly, the span testing was terminated for that condition. Children were given a short break and then the second test condition was administered.

Results

The computer recorded the display time for each card (how long the card took to count), the count answer (indicating counting accuracy), and the subsequent attempts at recall. Table 1 details the rate of counting cards at first and last positions and the time taken to work through the final card. Where a child miscounted the array, their answer was taken as the number of counted squares in deriving the count rate value.

Processing (card counting) speed. Given that the same ensemble of cards is counting in the short-final and long-final conditions, counting time across all card positions should be equivalent for these two conditions. The mean time to count each square was .69 and .68 s for the long-final card and short-final card conditions, respectively, a nonsignificant difference, $t(df = 66) = .39$.

Analysis then focused on whether the speed of counting targets varied as a function of whether the card was the first or last on a trial and whether the speed of counting the last card differed according to whether it was short or long. Analysis of variance on the counting rate, with age and card positions as factors, showed counting speed was significantly faster on the final card compared with the first $F(1,63) = 4.18$, $MS_e = .01$, $p < .05$ (see Table 1, top). There was also a significant increase in speed with age, $F(3,63) = 5.79$, $MS_e = .05$, $p < .01$, and an interaction between these effects, $F(1,63) = 4.76$, $MS_e = .01$, $p < .01$, with the increase in speed larger for younger children. Tukey HSD tests (corrected for unequal sample size; Kirk, 1968) indicated that the 6-year-olds counted significantly more slowly than 9- and 10-year-olds.

Further analysis of variance on the rate of counting each square on the final card showed

that while counting speed increased monotonically with age, $F(3,63) = 5.64$, $MS_e = .06$, $p < .01$, the counting speed on small and large cards was equivalent, $F(1,63) = .09$, $MS_e = .01$, and there was no interaction between these effects, $F(3,63) = .43$, $MS_e = .01$. Thus, children tackled each square on small-final and large-final cards at the same rate. This means, of course, that the overall time to count large cards was longer than the time to count short cards as seen in Table 1 (bottom), which was confirmed statistically, $F(1,63) = 244.2$, $MS_e = .37$, $p < .01$. Other effects were nonsignificant: age ($F(3,63) = 2.30$, $MS_e = 1.76$) and age by card type ($F(3,63) = .07$, $MS_e = .37$).

Processing (card counting) accuracy. The previous analysis does not take errors into account. However, in terms of the accuracy of counting, performance was extremely good, with only 2.2% of cards miscounted, and most of these errors reflected a failure to enumerate all the objects (a square was omitted from the count sequence). Analysis of errors is constrained by extreme floor effects, but a sign-test comparing the error rate per counted square showed no reliable difference between small and large final cards, $p > .05$, with 14 subjects making more errors on large arrays, five subjects making more errors on small arrays, and the remaining 48 subjects showing no difference. Similarly, there was no significant difference in the error rate per unit time spent counting (Wilcoxon test, $z = 1.25$, $p > .05$).

Counting span. Counting span was assessed in two ways. First, following Case et al. (1982) but requiring serial recall, span was measured as the maximum level at which recall was correct, with an additional 0.3 points if a single trial at the next level was also correct. Second, a more elaborate scoring procedure was adopted that takes account of the number of correct responses in a trial containing errors (for details, see Appendix A). Both measures produce a converging pattern of results, and only the latter score is detailed here. Analysis of variance, with age and final card size as factors, showed a reliable increase in counting span scores among older children, $F(3,63) = 29.1$, $MS_e = .79$, $p < .01$. Tukey HSD tests showed that the 9- and

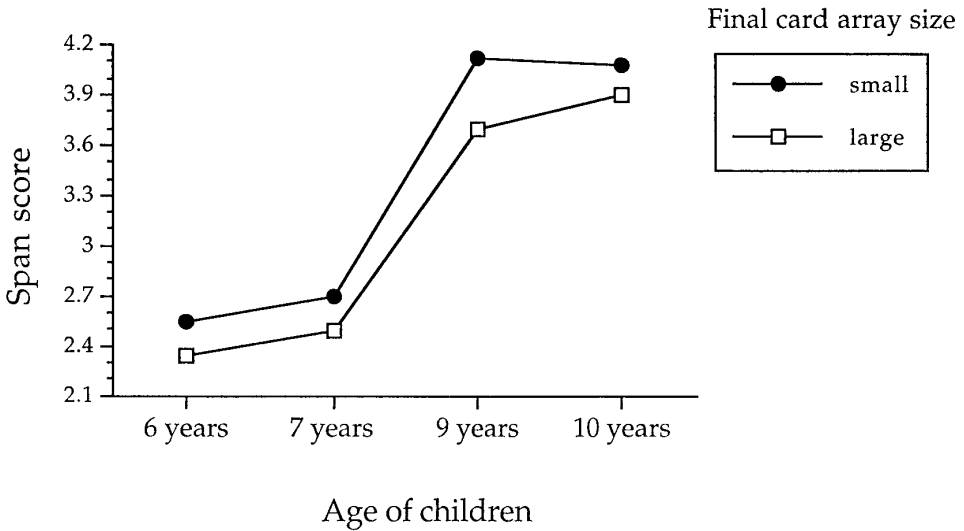


FIG. 3. The effect of manipulating card completion order on counting span in Experiment 1.

10-year-olds reached higher counting spans than both the 6- and 7-year-olds, although there were no significant differences within these groupings. There was also a significant effect of final card size, $F(1,63) = 7.56$, $MS_e = .28$, $p < .01$. That is, when the final array was large and increased the retention interval of card totals, there was a significant reduction in the counting span scores. This is illustrated in Fig. 3. There was no interaction between age and final card size, $F(3,63) = .49$, $MS_e = .28$.

Finally, we report the correlations involving measures of counting span, collapsing across final card size conditions to increase the stability of the measures. There was a significant correlation between counting span and counting speed (measured as the rate of counting each square), $r(65) = -.53$, $p < .01$. This correlation was still significant once the effect of age had been partialled out, $r(64) = -.34$, $p < .01$. Thus counting more rapidly was associated with higher counting span scores.

Summary. In addition to main effects of age in most analyses, results indicate that counting speed increased on later cards in a sequence. A large-final card took longer to count (and thereby increased the retention duration of the answers to previous counts) but did not affect the rate at which squares were enumerated, and so did not

indicate a change in the processing demand. And large arrays were not miscounted with disproportionate frequency. For memory span, there was a difference between final card condition, with smaller spans when processing time was experimentally extended. Further, large memory spans were associated with faster counting rates.

Discussion

The experiment provides converging evidence that memory decay is a significant determinant of children's working memory capacity, as measured by the counting span test. These findings are therefore highly concordant with those obtained by Towse & Hitch (1995) and the implication of data obtained by Halford et al. (1994). Here, results show that even quite subtle manipulations of memory decay parameters can affect children's spans. Only the numerosity of the first and last cards was varied in this study, and this temporal manipulation should be seen in the context of the overall task duration, which would be considerably larger. Even though the manipulation is rather fine-grained in this respect, it nevertheless has a reliable impact on span. Furthermore, there is no evidence that this temporal-based effect on counting span changes within the age range sampled here. While of course other variables

may influence the overall performance level of children, retention duration has a similar impact on different ages: there is no evidence for a developmental change in the way the counting span test is performed.

The findings are also noteworthy in that the present approach avoids the comparison of counting span in conditions where the sets of cards are different in nature, either by being more attentionally demanding or involving the counting of higher numbers. In this experiment, the ensemble of cards is the same, and so there is no reason to predict any change in span on the basis of standard resource-sharing accounts. The obtained effect of card completion order is therefore inconsistent with theories of working memory capacity that only consider processing intensity as a relevant variable. Also, the finding that children counted the final card on each trial more quickly than the first is not at all compatible with resource sharing models, as these predict that higher memory requirements at the end of card set would interfere with counting.

The reliable correlations between counting span and counting speed are interesting. In this study, we replicated the results of Case et al. (1982), who obtained significant correlations between counting speed and span, even when age had been partialled out. In contrast, Towse & Hitch (1995) failed to find a reliable partial correlation. A number of issues are relevant to this difference. The arrays in this study are homogeneous, in contrast to the use of cards with attentional demands and array numerosity systematically manipulated (Towse & Hitch, 1995). Plausibly, this homogeneity can be seen to facilitate the averaging of speed and span measures across experimental conditions and may yield more stable indices of performance. Furthermore, in this study, counting speed is measured at the time to count each square, a value which need not be affected by the array size. Additionally, the automation of the task allows more accurate measurement of counting speed on each card and exclusion of the intercard interval from the counting time.

EXPERIMENT 2

The previous study has shown the importance of processing duration in the determination of

counting span. This clearly places constraints on any theoretical approach to working memory tasks which stresses the role of resource sharing among children. However, the previous result might be thought to reveal more about counting span tests than working memory span tests generally. Consequently, children's performance is examined next with an operation span test (e.g., Turner & Engle, 1989), where again the completion order for a set of processing tasks is manipulated. Due to their relative lack of arithmetic sophistication, 6-year-olds were not included in the experimental design.

Method

Subjects and Design

Data are reported for 55 children in three age classes, attending a Surrey primary school. There was 17 8-year-olds (mean age 7.11, range 7.5 to 8.4), 18 10-year-olds (mean age 10.1, range 9.5 to 10.5), and 20 11-year-olds (mean age 10.10, range 10.5 to 11.5). All children completed two operation span tests with presentation order rotated across subjects. Children's performance with sets in which the final trial involved a (relatively) long arithmetic operation and a (relatively) short arithmetic operation is termed the effect of final card size and formed the within-subject manipulation in this experiment.

Procedure

The procedure largely followed Experiment 1, with all events controlled on an Archimedes A3000 microcomputer. Each "card" in the set comprised an arithmetic sum with the answer omitted: children read the arithmetic problem and then verbally reported the answer. They were not required to vocalize the operations and in fact did so infrequently. After completing all cards, children recalled the answers they had obtained. The arithmetic tasks always required a single digit answer and took one of the following forms:

$$\begin{aligned} A \pm B &= && \text{short (2-term) problems} \\ A \pm B \pm C &= && \text{intermediate problems} \\ A \pm B \pm C \pm D &= && \text{long(4-term) problems} \end{aligned}$$

where terms B, C, and D were either 1 or 0.

Therefore, examples of a short operation include “ $7 + 1 =$ ” and “ $8 + 0 =$ ” and a long operation “ $5 + 1 + 1 + 1 =$ ” and “ $8 + 0 + 1 - 1 =$.” Note that, for both short and long operations, the correct solutions are the same (i.e., in all the above examples, “8” is the correct answer). Therefore the set of values held in memory during the span tests do not differ for either short-final or long-final cards, and it is merely the processes used to derive these values that vary. Pilot work had shown errors in this arithmetic task sometimes involved a failure to notice a change in the mathematical operation to be performed (see last example above). To alert children to the requirements, displays were color-coded; digits and the equals sign appeared in yellow, plus signs in green, and minus signs in red. The computer randomly selected from a pool of arithmetic questions that led to a predetermined set of answers, forming the memory stimuli. As in the previous experiment, children began by attempting to remember answers from two cards, and the number of cards was gradually increased until recall was unsuccessful on at least two of three trials.

Results

Analyses are based on data from 55 children; two children were excluded since their error rate on the arithmetic operations was particularly high, and therefore it was not clear that they performed the processing component of the span task appropriately. The computer recorded response times and answers to the sums and the set of items which was subsequently recalled. Subjects took a mean of 4.77 s per card to respond in the long-final condition, and 4.75 s in the short-final condition, a nonsignificant difference, $t(df = 54) = .19$. In other words, any differences between final card conditions arise from the ordering within sets, since they required overall the same completion time.

Processing time. To determine whether children completed sums at the same rate through the processing phase of the task, analysis of variance compared calculation times, measured in seconds per sum, on first and last cards, with age and order of administering the span tasks as between-subjects factors. There was a strong

TABLE 2
Processing Times for Arithmetic Problems
in Experiment 2

	Age		
	8	10	11
Operation processing rate (seconds for each sum within a problem)			
First cards	1.94	1.54	1.35
Last cards	2.14	1.49	1.42
Processing time for short and long problems (in seconds)			
2-term (small) problems	4.33	2.89	3.03
4-term (large) problems	7.75	5.77	5.28

age effect due to quicker responses among older children, $F(2,49) = 14.4$, $MS_e = .30$, $p < .001$ (see Table 2, top). There was no significant difference in counting rate across presentation position, $F(1,49) = 1.66$, $MS_e = .13$. That is, subjects neither consistently speeded up nor slowed down across problems in a set. There was an interaction between age and testing order, $F(2,49) = 4.70$, $MS_e .30$, $P < .05$, with younger children responding more slowly overall when given the long-final, followed by short-final, card condition.¹ Other effects were not significant; order ($F(1,49) = .65$, $MS_e = .30$), age by position ($F(2,49) = .73$, $MS_e = .13$), order by position ($F(1,49) = .92$, $MS_e = .13$), order by age by position ($F(2,49) = .10$, $MS_e = .13$).

Analysis of variance was conducted on the time taken to answer short (two-term) and long (four-term) problems, with age and testing order as between-subjects factors and one subject excluded due to missing data (for means see Table 2, bottom). This confirmed that short problems produced much quicker responses than long problems, $F(1,48) = 163.6$, $MS_e = 1.29$, $p < .001$. There was also a clear effect of age with older children producing faster responses,

¹ Mean count speeds (for youngest to oldest children, respectively) were 1.81, 1.43, and 1.55 s when tested with a short-final card first, and 2.25, 1.55, and 1.22 s when tested with a long-final card first.

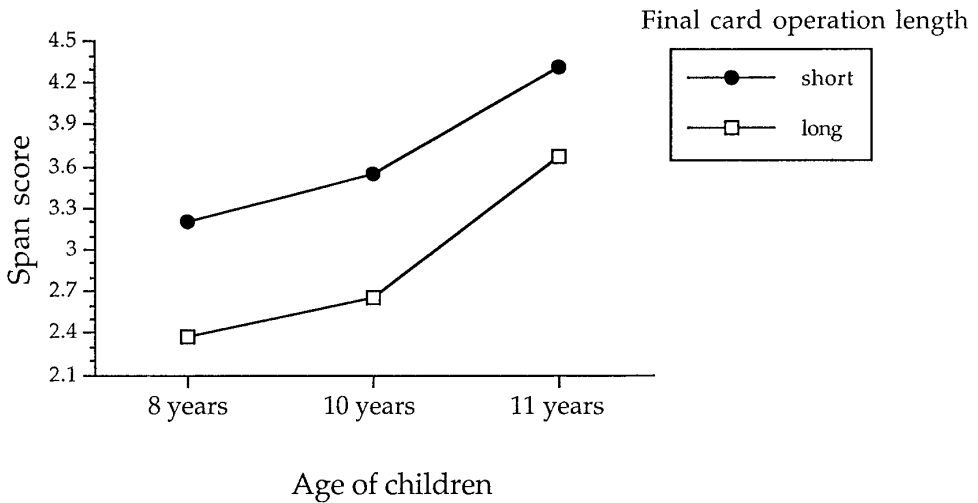


FIG. 4. Effect of manipulating card completion order on operation span in Experiment 2.

$F(2,48) = 9.00$, $MS_e = 4.12$, $p < .001$ (Tukey tests on the effect of age showed significant differences between 8- and 11-year-olds, and 8- and 10-year-olds, $p < .05$), along with a non-significant interaction trend between age and problem length, $F(2,48) = 2.55$, $MS_e = 4.12$, $p < .10$. Other effects were nonsignificant; order ($F(1,48) = .69$, $MS_e = 4.12$), age by problem length ($F(2,48) = 2.41$, $MS_e = 1.29$), order by problem length ($F(1,48) = .00$, $MS_e = 1.29$), order by age by problem length ($F(2,48) = .61$, $MS_e = 1.29$).

Processing accuracy. The previous section is based on all cards, ignoring errors. Examination of the accuracy of arithmetic operations showed that the great majority of answers were correct, with a mean of less than 1.3 errors overall for first and last cards on both test administrations (for more details, see Hutton, Towse, & Hitch, 1997). Due to floor effects, nonparametric analysis was employed. Comparing the effect of final card size, there tended to be more errors with long operations than with small operations. However, when miscalculations were expressed as an error rate (i.e., adjusting error scores on the basis of time spent deriving answers) there was no difference in the accuracy of short- and long-final cards, with adjusted mean error rates rounded to .82 errors per unit time per subject in both cases (Wilcoxon test, $z < 1$, $p > .10$).

Thus, for the final card, although there was a large difference in time to answer problems (thereby affecting processing duration), there was no consistent difference in the rate at which errors occurred (suggesting that processing intensity—"difficulty"—did not vary on the final card).

Operation span. Figure 4 illustrates the working memory operation spans for each age group. Analysis of variance on these spans with final card size, age, and testing order as factors confirmed that older children reached higher span levels, $F(2,49) = 11.1$, $MS_e = 1.34$, $p < .001$. Tukey tests showed significant span differences between 8- and 11-year-olds, and 10- and 11-year-olds, $p < .05$ or better. There was a strong effect of final card size, $F(1,49) = 35.5$, $MS_e = .43$, $p < .01$, with a short-final card leading to higher spans than a long-final card. Furthermore, this significant span advantage with short-final cards was also found when analyzing performance at each age group separately; for 8-year-olds, $F(1,15) = 22.5$, $MS_e = .25$, $p < .001$; 10-year-olds, $F(1,16) = 11.7$, $MS_e = .47$, $p < .01$, and 11-year-olds ($F(1,18) = 7.54$, $MS_e = .54$, $p = .01$). There was no interaction between age and final card size, $F(1,49) = .22$, $MS_e = .43$.

Although there was no overall effect of testing order (short-/long-final card condition

first), $F(1,49) = .03$, $MS_e = 1.34$, there was an interaction between final card size and order, $F(1,49) = 6.33$, $MS_e = .43$, $p < .05$. The impact of final card size was smaller when short-final operations were presented first (a difference of 0.39 items) than when long-final operations appeared first (a difference of 1.02 items). Order effects may arise because scores underrepresent actual skill levels where subjects err through incomplete understanding of task events, leading to span testing being curtailed early. Similarly there was also a just significant interaction between age and testing order, $F(2,49) = 3.31$, $MS_e = 1.34$, $p < .05$, arising from a stronger age improvement in spans when testing with long-final cards first.² The three-way interaction effect was not significant, $F(2,49) = .91$, $MS_e = .43$.

The average operation span achieved by each subject, together with a measure of the operation processing rate (seconds per arithmetic calculation) were calculated and correlated. There were significant correlations between operation processing rate and span, $r(53) = -.47$, $p < .001$, between age and span, $r(53) = .43$, $p < .001$, and between age and processing rate, $r(53) = -.59$, $p < .001$. The correlation between processing rate and operation span remained significant once the effect of age had been partialled out, $r(52) = -.29$, $p < .05$.

Summary of main results. Generally, there were age trends in all data sets, with improved performance among older children. Analyses of processing times confirmed that the manipulation of final card size affected the time spent answering the final card, but not the entire ensemble of cards. For the last card, errors occurred at the same rate for small and large operations. Operation span scores were reduced when the task completion order increased the retention interval for the memory items. Some analyses also showed interactions with order of testing (an incidental variable), but as these effects are not central they are not discussed further.

² The span level for short-final cards first was 3.19, 2.91, and 3.69 describing youngest to oldest children, respectively, with corresponding spans of 2.44, 3.15, and 4.31 for presentation of long-final cards first.

Discussion

Children's performance at an operation span task, in which a series of arithmetic problems are posed and the answers to each operation (or card) retained for subsequent recall, is similar to that on the counting span task. Once again, the order in which operations were carried out (whether the final card was quick or slow to complete) made a significant difference to the ability to remember the full complement of computed products. With a final card which took longer to complete, and hence imposed a greater retention interval for memory items, span was reduced. Indeed the strength of the effect in this experiment is such that operation spans varied significantly for each age group, as well as in the overall analysis. This possibly reflects the greater duration of long-final cards in the operation task (cf. Tables 1 and 2).

As in Experiment 1, the effect of manipulating retention interval on span was equivalent across age groups. This suggests that the temporal-based account of working memory span is not specific to a single age group, that the potency of temporal mechanisms does not vary significantly across age. The data also show a significant correlation between speed of processing and operation span (even after controlling for age effects); children who process the cards quickly reach higher spans. Such a finding gives additional (indirect) evidence that processing time provides an important constraint on the amount of information children can remember in the working memory span paradigm.

It is also noteworthy that the processing speed was equivalent on the initial and final cards (aggregating short and long operations). A resource sharing model of span would predict that as the processing task progressed (as children completed successive cards), response times would increase as a function of the memory load being carried concurrently. Insofar as processing speed did not vary across card positions, the within-trial analysis fails to provide evidence for resource sharing. This also reinforces the argument that simple resource-sharing models of working memory span are incomplete.

Importantly, then, the experiment replicates the effect of temporally based factors in working memory span found in Experiment 1. Thus, data show that the impact of processing duration is not restricted to counting span paradigms. Rather, the manipulation of retention time on working memory span appears to be a general task characteristic. Furthermore, the presence of retention time effects in each age group highlights not only the robustness of the effect in operation span tests, but also points to some type of task-switching mechanism occurring at different developmental levels of span performance.

EXPERIMENT 3

Reading span is another commonly used instrument for measuring working memory, a task where individuals read and comprehend a sentence and attempt to remember a text-relevant item. In this experiment, we presented both operation span and reading span tests to children. The administration of operation span allowed not only a replication of Experiment 2, but an opportunity to examine the similarities between working memory span tasks. For reading span, we chose to use a sentence completion task as the processing operation (see Leather & Henry, 1994; Siegel & Ryan, 1989). That is, a sentence was presented with the final word missing, and the task was to supply this sentence-terminal item. Since production of a missing word requires that the sentence be understood, a correct answer provides a good indication that processing requirements of reading and comprehension were met. Children subsequently recalled the set of missing words they had supplied, and their ability to recall items correctly yields a measure of reading span.

In addition to the working memory span tests, separate measures of processing speed were obtained, in the form of number comparison and identical picture tasks (Ekstrom, French, Harman, & Dermen, 1976). Since these tests have been used previously to estimate children's processing speed abilities (e.g., Kail & Park, 1994), they provide a somewhat more generic, or "general" measure of information-processing than is obtained from the task-specific perfor-

mance on working memory span (we use the term general speed as a label of convenience, however, since we do not wish to claim that these tests tap information-processing speed in any precise or pure way). These data, then, allow an estimation of the contribution to working memory span made by the on-line processing speed and by less specific, domain-independent factors.

Method

Subjects and Design

Data are reported from 65 children in three age classes, attending a Surrey primary school. There were 18 children in an 8-year-old group (mean age 8.4, ranging 7.10 to 8.9), 27 children in a 9-year-old group (mean age 9.3 ranging 8.10 to 9.8), and 20 children in a 10-year-old group (mean age 10.3, ranging 9.8 to 10.8). For administrative reasons, operation span data were available for only 62 of these children. Reading span and operation span (accompanied by picture matching and number comparison tests, respectively) were measured in separate sessions, with order counterbalanced. The completion order of the two forms of working memory tests (i.e., with short- and long-final card) was an incidental variable nested within the working memory task conditions.

Procedure

Processing speed tasks. In the number comparison task a sheet of paper was presented showing two columns made up of pairs of numbers (between 3 and 12 digits in length). These numbers were either identical or differed by a single digit. Children worked down the columns marking those number pairs that differed as quickly as possible. The number of correct responses in a 3 min interval (imposed to maintain task motivation) was recorded. In the identical pictures test, a target picture was presented along with a line up of five candidate pictures (one of which matched the target). Children marked a box underneath the matching target picture and completed a set of 24 picture matches as quickly as possible. The time taken to complete this task was recorded. There are

two versions of each speed test (Ekstrom et al., 1976): one form was chosen at random for each child, ensuring an equal distribution of tests.

Working memory tests. Operation and reading span were assessed using a Macintosh Powerbook 5300c computer, incorporating a high resolution, wide-angle TFT 10.25-in LCD color screen. Experimental responses were entered through an external keyboard—this allowed unobtrusive data entry and provided subjects with a continuous viewing of the computer screen. The operation span test was essentially as described for Experiment 2, with numerals now displayed in blue and the = symbol in black, and an essentially immediate presentation of a subsequent event following completion of a sum.

Procedure for the reading span tests corresponded to that for operation span, except for the nature of the processing task. A series of sentences was constructed in which the final word was missing, but where this word was relatively predetermined. Children read each sentence to themselves, although they were permitted to read aloud (which they did infrequently). Thus an example sentence is: “The magician waved his magic _____” where the subject would be expected to respond “wand.” A subset of sentences comprised two types—a short and long form—as in the two sentences: “The dog wagged his _____” (tail) “The dog was very happy and so he wagged his _____” (tail).

Both sentences involve the derivation of the same word (i.e., produce the same memory item) but due to sentence length they differ in reading time. The long sentences aimed to provide additional information to help specify the final word (or at least be neutral in this respect) and avoid unnecessary processing complexity. Note that sentences were not repeated across reading span trials or condition. Where the subject gave the expected response to a sentence, a single keystroke was used by the Experimenter to cue the next experimental event. When an alternative response was given, the Experimenter typed this in. However, after entering the first letter of this unexpected word, the screen presented the next experimental event

TABLE 3
Processing Times for Arithmetic Problems
in Experiment 3

	Age		
	8	9	10
Operation processing rate (seconds for each sum within a problem)			
First cards	1.79	1.49	1.48
Last cards	1.97	1.75	1.62
Processing time for short and long operations (in seconds)			
Short problems	3.81	3.28	3.09
Long problems	7.38	6.49	5.87

(and began timing this where appropriate) while allowing the Experimenter to continue typing the preceding response. Again it was the correspondence between supplied word and recalled word which formed the basis for assessing memory performance.

Results

Operation span. Subjects required a mean time of 4.95 s to respond to arithmetic operations across cards in the short-final condition and a mean time of 5.01 s in the long-final condition, a nonsignificant difference, $t(df = 61) = .42$. As in previous experiments, this confirms that varying the order of cards within a trial did not produce changes in the set considered as a whole.

To determine whether arithmetic solutions were derived at the same rate across the processing phase of the task, analysis of variance compared response times on the first card and the last card, with age and testing order as between-subjects factors (see Table 3, top), ignoring accuracy. There was a marginal effect of age, $F(2,56) = 2.72$, $MS_e = .42$, $p = .07$, and an interaction between age and testing order, $F(2,56) = 3.29$, $MS_e = .42$, $p < .05$ (the youngest age group responded 0.62 s more slowly when tested with a long-final card first, while the two older groups showed corresponding differences of .03 and $-.09$ s). There was a sig-

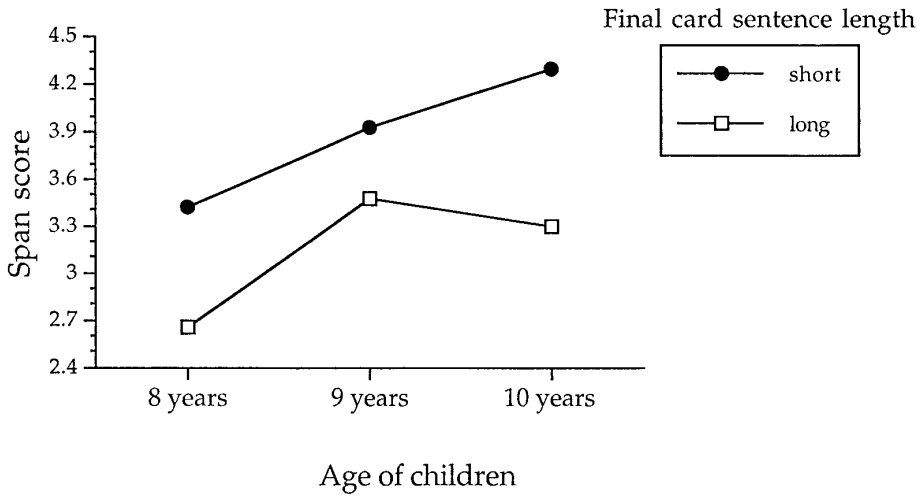


FIG. 5. Effect of manipulating card completion order on operation span in Experiment 3.

nificant slowing in response speed from first to last card, $F(1,56) = 17.8$, $MS_e = .07$, $p < .001$, and an interaction between this main effect and testing order, $F(1,56) = 8.30$, $MS_e = .07$, $p < .01$. When tested with long-final cards first there was a slowing in speed during the task (.33 s), an effect not found when processing short-final cards first (a corresponding difference of .03 s). Other effects were nonsignificant: order ($F(1,56) = 2.52$, $MS_e = .42$), age by position ($F(2,56) = .76$, $MS_e = .07$), order by age by position ($F(2,56) = 1.67$, $MS_e = .07$).

When analysis of variance was conducted on the time taken to answer short (two-term) and long (four-term) problems (see Table 3, bottom), there was a significant additional time requirement for long problems, $F(1,56) = 281.5$, $MS_e = 1.10$, $p < .001$. There was a significant improvement in processing speed with age, $F(2,56) = 3.32$, $MS_e = 3.52$, $p < .05$ and testing order, $F(1,56) = 4.55$, $MS_e = 3.52$, $p < .05$ as well as a borderline interaction between age and order, $F(2,56) = 3.17$, $MS_e = 3.52$, $p = .05$ (see comments above on the age by order interaction). There was also an interaction between order and problem length, $F(1,56) = 6.26$, $MS_e = 1.10$, $p < .05$. The age by problem length interaction was not significant, $F(1,56) = 1.38$, $MS_e = 1.10$, while there was a three-way interaction between age, order,

and problem length, $F(2,56) = 3.26$, $MS_e = 1.10$, $p < .05$. When presented first, the long-final card condition showed a larger age effect, and this was particularly marked for longer operations.³

Figure 5 describes the operation spans for each age group. Analysis of variance on span scores with final card size, age, and testing order as factors revealed a significant rise in span among older children, $F(2,56) = 4.99$, $MS_e = 1.27$, $p = .01$. Tukey tests showed that the spans of the 8-year-old group were significantly smaller than for the 9- and 10-year-olds ($p < .05$). There was also a significant effect of final card size, with a longer problem and larger memory retention intervals reducing operation span, $F(1,56) = 27.8$, $MS_e = .60$, $p < .001$. This was also true for each age group—for 8 year-olds, $F(1,16) = 19.8$, $MS_e = .27$, $p < .001$, 9-year-olds, $F(1,23) = 5.82$, $MS_e = .48$, $p < .05$, and for 10-year-olds, $F(1,17) = 8.63$, $MS_e = 1.08$, $p < .01$. Other effects were nonsignificant: order ($F(1,56) = .95$, $MS_e = 1.27$), order

³ Thus, for short problems, the computation time for youngest to oldest children with a short-final card first was 3.41, 3.24, and 3.16 s, and for a long-final card first 4.21, 3.33, and 3.03 s. For the long problems, corresponding completion times were 5.86, 6.06, and 6.05 for a short-final card first, and 8.89, 7.03, and 6.56 s for a long-final card first.

by age ($F(2,56) = 1.15$, $MS_e = 1.27$), order by final card ($F(1,56) = .73$, $MS_e = .60$), age by final card ($F(2,56) = 1.19$, $MS_e = .60$), order by age by final card ($F(2,56) = .71$, $MS_e = .60$).

Once again children were largely successful at the arithmetic operations, making an average of less than one error on the final card of each condition throughout testing (see Hutton et al., 1997). Using a Wilcoxon test, comparison of the rate at which errors were made on the last card showed no significant difference between short and long cards, $z = .43$, $p > .10$, and therefore both short and long cards appeared to involve the same processing intensity.

Summary. Age effects were obtained across analysis. Operation speed declined from the beginning to the end of a trial, for one presentation order only. Long problems were slower to complete, but proportionately, were not more difficult to answer. Operation spans were smaller when time before retrieval was increased, for overall data and for each age group.

Reading span. Performance on reading span was subjected to the same battery of analytic tests as operation span. As an initial check on the experimental manipulations, the time required to complete problems was compared on the two experimental conditions. Subjects took a mean of 6.94 s to provide answers to sentences in the short-final condition and required a mean of 6.83 s in the long-final condition, a nonreliable difference, $t(df = 64) = .32$. As with previous analyses, then, manipulation of card order within a set did not affect overall response times.

Analysis of variance on the reading times for all first and last cards, ignoring accuracy (for means, see Table 4, top), with age and testing order as between-subjects factors, revealed a significant increase in reading speed with age, $F(2,59) = 5.81$, $MS_e = 25.1$, $p < .01$. There were no other significant effects: no significant difference between speed on the first and last card, $F(1,59) = 1.91$, $MS_e = 4.08$, or effects of order ($F(1,59) = 2.13$, $MS_e = 25.1$), order by age ($F(1,59) = .55$, $MS_e = 25.1$), order by position ($F(1,59) = .71$, $MS_e = 4.08$), age by position ($F(2,59) = .64$, $MS_e = 4.08$), order by age by position ($F(2,59) = .65$, $MS_e = 4.08$).

TABLE 4			
Processing Times for Sentences in Experiment 3			
	Age		
	8	9	10
Sentence processing time at the beginning and end of a trial (seconds)			
First cards	8.56	7.01	4.99
Last cards	9.06	8.00	5.00
Overall processing time for short and long sentences (in seconds)			
Short sentences	6.97	6.61	3.87
Long sentences	10.65	8.40	6.12

Furthermore, comparing the reading times for short and long sentences, analysis of variance confirmed that words were supplied to long sentences more slowly, $F(1,59) = 55.4$, $MS_e = 3.80$, $p < .001$ (see Table 4, bottom).

For reading span, there was no significant improvement with age, $F(2,59) = 2.30$, $MS_e = 1.05$. There was, however, a clear difference between reading spans on short- and long-final cards, $F(1,59) = 28.6$, $MS_e = .19$, $p < .001$, see Fig. 6. Thus, a long retention interval reduced reading span scores. This was also true for the 8-year-olds alone, $F(1,16) = 21.9$, $MS_e = .17$, $p < .01$, 9-year-olds, $F(1,25) = 5.38$, $MS_e = .23$, $p < .05$, and 10-year-olds, $F(1,18) = 5.72$, $MS_e = .16$, $p < .05$. Other effects were not significant: order ($F(1,59) = .00$, $MS_e = 1.05$), order by age ($F(2,59) = 2.60$, $MS_e = 1.05$), order by final card ($F(1,59) = 1.74$, $MS_e = .19$), age by final card ($F(2,59) = 2.06$, $MS_e = .19$), order by age by final card ($F(2,59) = .67$, $MS_e = .19$).

Analysis of errors on the reading task is less straightforward than for the other working memory tasks. Children’s modal response was to give the expected word (over 90% were target words). Of the nontarget responses, most were semantically related, and formed more-or-less appropriate productions, showing clear evidence of semantic comprehension, (e.g., “A joke makes me *giggle*” rather than the target *laugh*). Occasionally, responses were less appropriate, (e.g., “We live on the planet *Mars*”

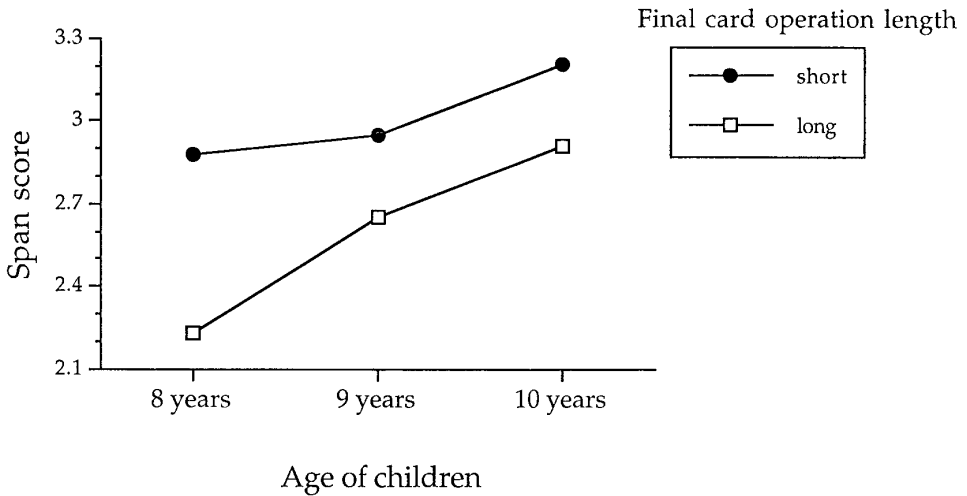


FIG. 6. Effect of manipulating card completion order on reading span in Experiment 3.

rather than *Earth*). Taking responses which were neither expected, nor subjectively plausible, there were eight errors made from short sentences and 10 errors from long sentences (further description of all sentence responses is provided in Towse, Hutton, and Hitch, 1997). Clearly, the two sentence lengths were equated in terms of the difficulty of producing target, or close-to-target, words.

Summary. Although reading speed was positively associated with age, memory spans did not increase significantly with age. Reading times were equivalent at the beginning and end of a trial, and sentence completion responses were equally appropriate for short and long sentences. Reading span was reduced with longer retention delays, at each age group.

Relationship between measures. A series of zero-order correlations were computed between chronological age, general measures of processing speed obtained from number comparison and picture matching tests (in both cases processing rate was expressed as time per correct question answered; see Table 5), reading span and speed, and operation span and speed. For these working memory measures, composite scores were obtained by averaging standardized values from small-final and large-final experimental conditions. These correlations are displayed in Table 6. As in Experiment 2, there

was a significant correlation between operation speed and operation span, a relationship which remained significant once age had been partialled out, $r(58) = .31, p < .01$. Similarly, the correlation between reading speed and reading span was not eliminated by partialling out the effect of age, $r(61) = -.51, p < .01$. In contrast, the correlation between the two measures of general processing speed (number comparison and picture matching) was removed by partialling out age, $r(59) = .24, .05 < p < .01$.

As in previous studies, then, there is a relationship between the speed of performing cognitive operations and the ability to hold the products of this cognitive work. The present data, however, also permit a further question to be addressed; given that task speed predicts span, is there an independent contribution to

TABLE 5

Processing Times for Identical Picture and Digit Comparison Test

	Age		
	8	9	10
Seconds per correct problem answered			
Identical pictures	4.72	3.83	3.56
Digit comparison	9.36	8.38	7.86

TABLE 6
Correlations between Performance Measures in Experiment 3

	1	2	3	4	5	6	7
1. Operation span							
2. Operation speed	-.40**						
3. Reading span	.37**	-.20					
4. Reading speed	-.42**	.39**	-.54**				
5. Picture matching	-.49**	.40**	-.29*	.35**			
6. Number comparison	-.38**	.37**	.19	.27*	.32*		
7. Age	.34**	-.38**	.26*	-.47**	-.47**	-.24	

* $p < .05$.
** $p < .01$ or better.

span performance from the more generic measures of processing speed? A composite, general speed measure was obtained by averaging standardized number comparison and picture matching data. Then, sequential multiple regressions were performed using working memory span as the dependent variable. (With the use of a $p < .001$ criterion for Mahalanobis distance, no multivariate outliers were found for the reading span analysis but one outlier was established for operation span analysis. Winsorizing operation speed data for this case by curtailing the value to three standard deviations from the mean provided an acceptable rescaling function.)

With operation span as the dependent variable, in the first step age was entered into the regression equation, indicating that older children reached higher spans, $R^2 = .12$, adjusted $R^2 = .10$, $F(1,59) = 7.78$, $p < .01$. Operation speed scores were entered in the second step, which accounted for significant further variance over the above age, $\Delta R^2 = .08$, $F(1,58) = 5.84$, $p < .05$. In the third step, the general speed scores were entered, and this variable too added significant unique prediction to span, $\Delta R^2 = .12$, $F(1,57) = 10.2$, $p < .01$. Further analyses showed that both operation speed, $R^2 = .15$, adjusted $R^2 = .14$, $F(1,59) = 10.7$, $p < .01$, and then general speed, $\Delta R^2 = .16$, $F(1,58) = 13.6$, $p < .01$, independently predicted operation span, but age failed to add significantly to this prediction, $\Delta R^2 < .01$, $F < 1$. When entered after age and general

speed, however, operation speed did not significantly explain further variance in operation span, $\Delta R^2 = .02$, $F(1,57) = 1.82$.

A corresponding analysis was then performed for reading span scores. Again, older children produced significantly higher spans, $R^2 = .07$, adjusted $R^2 = .05$, $F(1,62) = 4.48$, $p < .05$. Reading speed was entered at the second step and accounted for significant additional variance, $\Delta R^2 = .24$, $F(1,61) = 21.2$, $p < .01$. Entered as a third step, the general speed measure did not predict span significantly, $\Delta R^2 = .01$, $F(1,60) = 1.07$. Reading speed also predicted reading span, $R^2 = .31$, adjusted $R^2 = .30$, $F(1,62) = 27.6$, $p < .01$, but neither general speed, $\Delta R^2 = .01$, $F < 1$, nor age, $\Delta R^2 < .01$, $F < 1$, provided any further prediction when reading span scores were entered first. Reading speed continued to predict significant variance on reading span after accounting for age and general speed measures, $\Delta R^2 = .21$, $F(1,60) = 18.5$, $p < .01$.

Discussion

The operation span performance in this experiment replicates that from Experiment 2 in several important respects. The manipulation of card completion order was successful in varying the retention demands of the arithmetic operation span task (while preserving equivalent task difficulty levels) and this impacted on the overall operation spans achieved by children, as well as the operation spans for each age group. Similarly, in completing a version of the reading

span tests in which children read a sentence to supply a missing, sentence-terminal word, extending the retention interval for the words degraded the quality of later memory retrieval. Thus the impact of retention interval is confirmed in a nonnumerical processing domain and with an unlimited stimulus pool (i.e., sampling for memory items across trials without replacement; see LaPointe & Engle, 1990, for a consideration of this issue in the context of working memory tasks).

In this experiment, operation span tests showed a significant slowing in performance speed comparing first and last cards (though note this was not mirrored in reading span tests, nor found in Experiment 2). One interpretation of this finding is that there was resource sharing between processing and storage components of the task; i.e., as the working memory load increased, so the processing speed was degraded. However, there is also an alternative explanation. That is, despite instructions not to engage in rehearsal of memory items during the arithmetic processing, children may have engaged in the numerical task only after they had attempted some rehearsal of earlier items. Obviously, this would increase response times on the final card. However, since this effect appeared only for one of the two working memory tests, for one presentation order of card conditions, and was not found with the same task in the previous experiment, we suggest that caution is required in its interpretation. Additionally, while certainly intriguing, note that it does not detract from the strength of evidence for sensitivity of span to retention interval.

The inclusion of separate measures of cognitive speed provided an opportunity to compare the role of task-specific processing speed (i.e., the speed component considered in earlier experiments) with any general, overarching processing speed factor. The picture comparison and number comparison tests were used for this purpose (Kail & Park, 1994), though note that in doing so, a numerical task is incorporated into a task which is contrasted with operation speed. The results provide guarded additional support for the conclusion that the speed-span relationship is driven by task-specific speed processes.

The regression models showed that general speed measures did not significantly explain reading span scores over and above reading speed scores. Similarly, although age was able to predict reading span scores when entered first, it did not uniquely contribute to the explanation of span after accounting for reading speed. For operation span, the results were slightly different, for general speed scores accounted for significant variance over and above the effect of age and operation speed. Age, however, did not predict operation span after accounting for operation speed and general speed. Tentatively, then, it would appear that the retention of numerals is influenced by factors external to the task configuration itself.

The picture that emerges from the regressions is one in which speed at the relevant processing operation is important for explaining working memory span. A conclusion of this sort is certainly consistent with the general account of working memory span developed here, since forgetting in working memory tasks is considered to occur while the subject is engaged in a processing task. Therefore, it is the speed of the on-line processing task which is likely to be informative. Notwithstanding this particular interpretation, and noting that more generic speed measures added significantly to prediction of operation span over and above operation speed, it should be apparent that the distinction between specific and general indices of processing speed, like the distinction between *g* and specific intellectual abilities, is not easy to determine, especially without complete precision in the selection of general tests.

GENERAL DISCUSSION

The present series of experiments confirms that tests of working memory span in children, namely counting span, operation span, and reading span, reflect common properties. That commonality, as uncovered here, lies in the sensitivity to the temporal dynamics of the task. In general, then, the probability of remembering the products of processing varies as a function of the time for which such products must be maintained in order to complete the task. Such an account clearly promotes the idea that an

important variable underlying memory development is the change in task processing speed. As Hitch and Towse (1995) pointed out, articulation speed provides a substantial (though not complete) explanation for development in standard short-term memory tests such as digit span. Likewise, the present data suggest that nonarticulatory processing speed can be used to account, at least partially, for changes in the performance of active, working memory tasks.

Standard resource sharing models have no way of explaining the dependence of span on the temporal dynamics of the task given that the total amount of processing was held constant across conditions. Specifically, such models have no basis for predicting changes in span when there is no change in the intensity of information processing or in the volume of processing elements.

Resource sharing models can also be evaluated in terms of the further prediction that processing should be impaired when concurrent storage load is high. Given that processing errors were infrequent, this prediction was tested by comparing the speed of completing first cards (zero memory load) and last cards (high memory load). Results showed that last cards were counted faster than first cards in counting span (Experiment 1), there was no significant difference for reading span (Experiment 3), and there was none for operation span in Experiment 2. Results for operation span in Experiment 3 complicate the picture in that while there was no difference for children tested in one administration order, last cards were processed slower than first cards for the reverse order. In the absence of consistent interference effects, the results provide no support for resource sharing models. In contrast, task-switching models are consistent with the most typical finding of no disruptive effect of concurrent storage load on processing operations. As described earlier, such models assume that children adopt a strategy such that either processing and storage do not compete at the same time in working memory, or alternatively, that storage and processing draw on separate, dedicated systems that do

not interfere (see, for example, Towse & Hitch, 1997).

It is important to emphasize that while supporting the task-switching interpretation, logically the present results cannot rule out the possibility of resource sharing in other situations or rule out resource sharing in adults. Further, the results do not pinpoint the mechanism by which products of information-processing operations are forgotten. As previously outlined, temporal decay or interference (e.g., the "acid bath" account of forgetting; Posner & Konick, 1966) may be responsible. Yet another possibility is that the forgetting we have observed is a function of the relative distinctiveness of increasing vs decreasing temporal intervals in memory (Neath & Crowder, 1990). Provisionally, we are inclined toward a decay interpretation, given the observation that working memory span was related to the speed at which processing was executed in all three experiments. There is no obvious reason why this would be the case if forgetting was based on either interference or distinctiveness.

To explicate the mechanisms responsible for children's working memory capacity, we have focused on children between approximately 6 and 10 years of age. Consequently, it clearly remains an open and important question as to whether fundamentally different processes are involved in working memory among older children and adults. Certainly, studies of working memory in adults have not supported the argument that task processing speed explains individual differences in working memory capacity (Conway & Engle, 1996; Engle et al., 1992). Such findings might be taken to imply simply that individual differences in capacity are to be distinguished from developmental differences in capacity (see, for example, Engle, 1996). An alternative, more radical possibility, is that adults may engage in resource sharing strategies, while children do not do so (or cannot do so); there may be a developmental progression away from task-switching in working memory span tasks.

We note that there was no evidence in our experiments that such development proceeds before the age of 10, as it was consistently the case

that all children were equally sensitive to the temporal structure of the span tasks. However, it is difficult to interpret this lack of interaction between age and performance with complete confidence as older children generally reached higher span levels and completed the tasks more quickly than younger children. Thus, their higher memory loads might be expected to make them more sensitive to delay, while on the other hand, their faster processing speed would reduce the temporal difference between experimental conditions. Added to these difficulties of interpretation, it may be that there is a nonlinear effect of time such that small temporal differences exert a greater importance at short overall delays (Peterson & Peterson, 1959). Therefore, this age-invariant aspect of working memory span data is not necessarily inconsistent with temporal explanations of performance, though it certainly points out inevitable unresolved issues.

In light of rather different datasets, Kail (1995) commented that many cognitive tasks are speed-limited; cognitive operations, be they mathematical computations or semantic inferences, must take place within a finite window of time or else they fail. Consideration of speed for memory processes is particularly pertinent since these cognitive operations often involve a set of subgoals, the products of which must be remembered if the final representation is to be appropriate (e.g., Adams & Hitch, 1997). The present series of experiments illustrates how rapid processing can be important for successful information retention. And from a theoretical perspective, the data emphasize the need to incorporate temporally based forgetting within models of working memory. The retention characteristics of counting span, operation span, and reading span tasks seem too great to be subserved primarily by a system such as the phonological loop within Baddeley's (1986) working memory architecture (especially since rehearsal was explicitly discouraged throughout), given that this loop system is subject to substantial information degradation within 2 s or so. Consequently, within the standard model the explanatory burden must be carried within the "central executive"

system (for which there is increasing evidence of fractionated functions; Baddeley, 1996; Lehto, 1996; Towse, 1998).

The current conceptualization of working memory (e.g., Baddeley, 1992; 1996) does not offer a temporal framework within which to explain "central" working memory capacity phenomena (i.e., as part of active memory tasks). We see little reason, however, why the model cannot accommodate such an approach. The current data sets emphasize that subjects can remember (limited amounts of) information for periods longer than the postulated holding capacity of the phonological loop. However, the integrity of the memory representations clearly becomes degraded over periods of time. Together with evidence from related studies (Halford et al., 1994; Towse & Hitch, 1995), one might also conclude that, among children, this memory substrate is not affected by the processing demands of the concurrent activity making up the task. While this can be seen to narrow down the viable theoretical positions outlined earlier, the data do not fully constrain the type of task-switching mechanisms which may be postulated. For example, although there is no evidence in the present findings to support the notion of a sequentially functioning system common to both processing and memory operations, it is not disconfirmed by present data either.

In conclusion, the argument that we draw on the basis of the present results is that retention duration is an important and general property of working memory span tests in children, a factor which may in some circumstances be more important than resource sharing. It does not follow, of course, that memory decay alone can account for working memory span performance (a point made previously; Towse & Hitch, 1995, p. 120). There are many other studies which indicate the range of important developmental changes taking place at this time—the emergence of "good information processing" being just one recent candidate (see Pressley, 1995). Clearly, working memory span development will reflect a combination of factors, and it would be misleading to identify processing speed as the *only* issue at stake. Nonetheless,

insofar as retention duration is important to working memory span, and processing speed improves over age, then changes in the profile of memory decay consequent upon faster cognition provide one important explanation for early memory development.

APPENDIX A

Counting Span: Criteria for Scoring

1. Span is initially scored as the highest level for which the majority of answers were correct.

2. If one of the three trials at the next level is correct, since this represents "half-way" toward the next span level, an additional 0.5 marks are given.

3. The number of correct answers (correct item in the correct serial position) in each remaining trial is divided by the number of items to be recalled. The mean proportion of correct recalls for the incorrect trials is then derived. This value is multiplied by 0.5 and the product added to the total score obtained from procedures 1 and 2.

3a. Transposition of two adjacent items ("7...4" recalled as "4...7") is scored as being one correct answer out of two recall items.

3b. Where one answer is omitted completely but subsequent items are correctly recalled ("7...6...5...4...3" recalled as "7...5...4...3") the subsequent items can be scored as correct, but two marks are deducted. The above example, therefore, would be viewed as 2 out of 5 (1 for the "7", 3 for the "5...4...3" and -2 for the omission).

Note. Procedure 3b is applicable only when a positive end value is obtained (it is not used in cases where there are two or less subsequent items). Procedure 3b can only be applied once to a recall string and procedure 3a and 3b cannot be used together ("7...6...5...4...3" recalled as "7...4...5...3" would be given a mark of only 1 out of 5, for the correct recall of "7").

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